



UNITED STATES
NUCLEAR REGULATORY COMMISSION
REGION II
101 MARIETTA STREET, N.W.
ATLANTA, GEORGIA 30303

MAR 12 1985

Report No.: 50-369/85-11

Licensee: Duke Power Company
422 South Church Street
Charlotte, NC 28242

Docket No.: 50-369

License No.: NPF-9

Facility Name: McGuire 1

Inspection Conducted: February 26-28, 1985

Inspector: D. M. Montgomery for
W. B. Gloersen

3-8-85
Date Signed

Approved by: D. M. Montgomery
D. M. Montgomery, Section Chief
Division of Radiation Safety and Safeguards

3-8-85
Date Signed

SUMMARY

Scope: This special, announced inspection entailed 18 inspector-hours on site investigating the radioactive spent resin release by way of the Unit 1 ventilation stack.

Results: One violation was identified - failure to follow operating procedure OP/O/B/6200/32 which resulted in the release of radioactive spent resin.

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REPORT DETAILS

1. Persons Contacted

Licensee Employees

- *G. Cage, Superintendent of Operations
- *D. Rains, Superintendent of Maintenance
- *J. Foster, Station Health Physicist
- *T. Keane, Corporate Health Physicist
- *T. Wall, Radwaste Chemical Coordinator
- R. Propst, Radwaste Engineering Processing Supervisor
- D. Mendezoff, Licensing Engineer
- *N. McCraw, Compliance Engineer
- *R. Johansen, Testing and Surveillance Engineer

Other licensee employees contacted included technicians and operators.

NRC Resident Inspectors

- *W. Orders, Senior Resident Inspector

*Attended exit interview

2. Exit Interview

The inspection scope and findings were summarized on February 28, 1985, with those persons indicated in paragraph 1 above. The inspector described the violation concerning the failure to place the Auxiliary Building Exhaust System in the "filtered mode" as required by radwaste operating procedure OP/O/B/6200/32 (Paragraph 3). One new inspector followup item was opened concerning the completion of a surveillance and testing program for the non-safety grade air filtration systems (Paragraph 3). Licensee representatives acknowledged the violation and the inspector's comments. The licensee did not identify as proprietary any of the materials provided to or reviewed by the inspector during this inspection.

3. Investigation of the Release of Radioactive Spent Resin Beads via the Unit 1 Vent (93702)

On February 20, 1985, the licensee discovered radioactive spent resin beads on the roof of the Auxiliary Building. The discovery was made after it was determined that an individual, who was repairing the Unit 1 "doghouse" roof and had tar on the soles of his shoes, stepped on some resin beads lying on the walkway of the Auxiliary Building roof and subsequently set off the portal monitor alarm at the health physics control point. The licensee determined (as explained below) that the resin was released from the Unit 1 ventilation stack and spread in a north northeast direction from that point. Resin was also identified in the following areas:

- Unit 1 Interior Doghouse Roof
- Unit 1 Fuel Building Roof
- Unit 1 Reactor Building Roof
- adjacent yard within 50 feet of the Auxiliary Building

A description of the incident and cause is given below. The release of the radioactive spent resin beads was believed to have occurred some time after 1:00 p.m., on February 19, 1985, during a routine transfer of resin from the Spent Resin Storage Tank (SRST) to the Resin Batching Tank (RBT). During a normal transfer process, radwaste operators visually monitor three "portholes" positioned at various levels on the RBT by using a video camera and television monitor. The transfer process is stopped when the water level reaches the third porthole which is approximately at the same level as the overflow line of the RBT. The inspector examined the water level television monitoring system and noted that it would be difficult to determine water level in the RBT once the water level went above the porthole. During the resin transfer on February 19, 1985, the radwaste operators may have allowed the water level to go slightly above the normal termination level on the third porthole. As the resin slurry mixture entered the RBT near the ventilation line located at the top of the tank, a possible "coning" effect caused some of the resin mixture to enter the ventilation line. The RBT is vented to the Unit 1 Auxiliary Building filtered exhaust. The licensee later determined that the RBT Overflow line was occluded with resin; however, the exact locations of the occlusions could not be determined. The resin eventually traveled through the Unit 1 Auxiliary Building Ventilation System, bypassing the filter train, and after drying out became airborne due to the high air flow and equipment vibration. The licensee reviewed the meteorological strip chart records and determined that the wind direction was generally from the SSW to SSE quadrants at approximately 10 mph at the approximate time of the release. Sampling and survey results confirmed that the resin was carried out of the Unit 1 ventilation stack and spread in the NNE direction from the release point.

A similar incident occurred in August 1984. In that incident, the Resin Batching Tank overflowed and an investigation by the licensee showed that approximately $1/3 \text{ ft}^3$ of resin collected on the prefilter side of the Unit 1 Auxiliary Building Exhaust System. Since the filtered exhaust is operated normally in a "bypass" mode, the licensee revised operating procedure OP/O/B/6200/32 "Radwaste Procedure for the Nuclear Solid Waste System Disposal Operation" so that the exhaust system would be placed in "filtered exhaust" mode whenever the resin transfer process is performed to prevent the possible release of resin through the exhaust.

During the resin transfer process on February 19, 1985, radwaste operators failed to notify the control room and request that the Unit 1 Auxiliary Building Exhaust System be placed in the filtered mode according to OP/O/B/6200/32, step 4.8.1.6. The inspector discussed the procedure with radwaste operators and determined that the procedure had been performed successfully by individuals operating the system since the procedure had been revised. On February 19, 1985, two individuals, instead of one,

performed the operating procedure and a lack of communication between the two operators resulted in the failure to place the Auxiliary Building Exhaust System in the "filtered mode." The inspector reviewed OP/O/B/6200/32 (Rev. 12, 1/8/85) and noted that there were neither sign-off sheets, nor verification steps to assure that the procedure would be implemented properly. The failure to place the Auxiliary Building Exhaust System in the "filtered mode" was the result of both an operational error and procedural inadequacy. The inspector informed cognizant licensee representatives that the failure to follow OP/O/B/6200/32 was a violation of Technical Specification 6.8.1 which requires procedures to be established, implemented and maintained. (50-369/85-11-01).

The inspector discussed with licensee representatives the various engineering modifications to the RBT ventilation line to prevent the resin from entering the ventilation system. Some of the options included: (1) routing a new ventilation line to an exhaust header downcomer that is connected to the Auxiliary Building Exhaust System but where the flow rate is much lower than where the RBT ventilation line is connected presently; (2) placing a resin screen over the existing ventilation line so that any inadvertent overflow would pass just water out the ventilation lines; and (3) routing a new ventilation line to the mixing and settling tank which is vented via Auxiliary Building Exhaust System. Presently, the licensee has placed the Auxiliary Building Exhaust System in the filtered mode and has cut and taped the end of the ventilation line until the problem has been resolved.

In addition, the inspector discussed/reviewed the licensee's clean up and environmental assessment of the resin release. The licensee determined that approximately one cubic foot of resin was released. The Unit 1 Reactor Building and Fuel Handling Building roofs have been generally decontaminated except for a few isolated "fixed" spots with greater than 5000 dpm/100 cm². The hard and concrete surfaces on the Auxiliary Building roof have been cleaned except for some "fixed" spots and any resin that has settled below the relatively large stones that are part of the roof. The licensee will continue to evaluate and determine a feasible method to either remove or cover the isolated fixed spots and the resin that has settled below the large-stoned Auxiliary Building roof. In addition, the licensee is monitoring the roof drain system in the event that rain would wash the resin through the drain. The existing level of contamination was low and posed no exposure hazard to personnel. The licensee has posted the contaminated areas. The contamination has been limited to areas within the protected area. The yard adjacent to the Auxiliary Building and yard drain have been decontaminated. The licensee collected and analyzed approximately 35 soil and/or sediment samples and determined that the nuclides found in a few of the samples were attributed to the nuclides contained in the resin beds. In general, the activity levels found in the samples were low. The highest level of activity was found in a soil sample collected immediately outside the protected area fence which contained amounts of Co-60 and Co-58 on the order of 10⁻³ μ Ci/Kg. Adjacent sample locations contained amounts of Mn-54, Co-58, Co-60, Cs-134, and Cs-137 on the order of 10⁻⁴ μ Ci/Kg.

The inspector reviewed the licensee's calculation of the estimated offsite dose as a result of the resin bead release. The licensee used resin size distribution information, resin density, release volume, and actual meteorology (wind speed and direction) to estimate the fraction of resin potentially transported to the exclusion area boundary. The dose estimate was based on the isotopic analysis and specific activity of the resin beads using the Offsite Dose Calculation Manual (ODCM). The land use census information was used to determine what dose pathways should be considered given the meteorological conditions that existed during the resin release. The licensee informed the inspector that an in-house formal report would be written to document the calculations.

In addition to investigating the resin bead release, the inspector reviewed the licensee's response to IE Information Notice No. 82-43: Deficiencies in LWR Air Filtration/Ventilation Systems. The information notice dealt with the need of surveillance and testing program for non-safety grade air filtration systems. The licensee was in the process of developing a surveillance program for the non-safety related systems including the Technical Support Center filter system, the air addition system in containment, count room supply filter system, lower containment cleanup system, and the incore instrument purge system. The inspector informed licensee representatives that the completion of the surveillance and testing program for the non-safety grade air filtration systems will be reviewed during a future inspection (50-369/85-11-02).

Finding: Violation - failure to follow operating procedure for transfer of spent resin.